**Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)**

On

# Smart Contract-Driven Energy Trading Mechanism Using Blockchain

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# Introduction

- Traditional energy systems operate on centralized grids where electricity flows from power plants to consumers.
- Such systems suffer from major inefficiencies like high transmission losses and lack of transparency.
- Operational costs increase due to involvement of intermediaries such as utilities and brokers.
- With renewable energy adoption, users now act as prosumers (both producers and consumers).
- This shift demands a decentralized and flexible energy trading mechanism.

# Problem Statement

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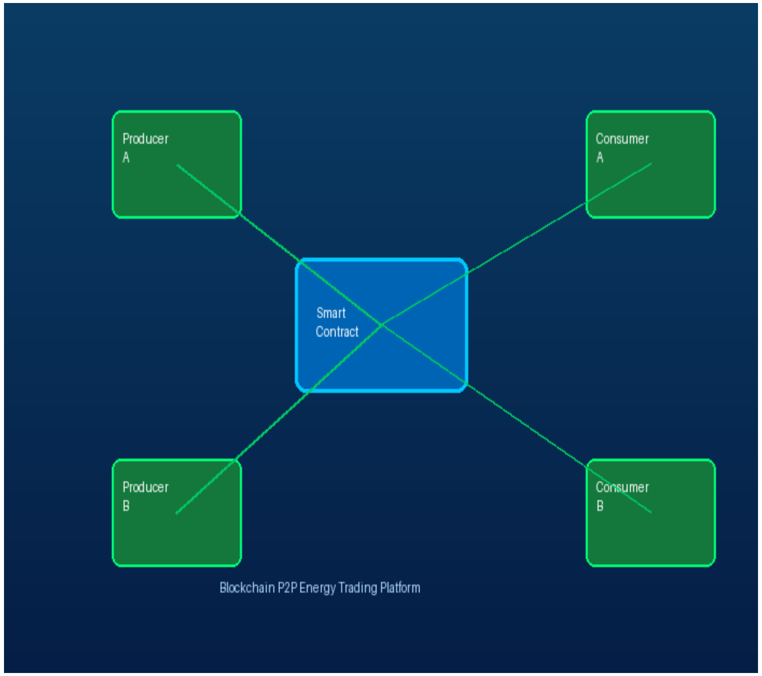
- Current systems do not support direct peer-to-peer energy exchange.
- Consumers have limited control over pricing and energy sourcing.
- Centralized control creates a single point of failure and security risk.
- Settlement processes are slow, often taking hours or days.
- Lack of trust and transparency reduces system efficiency.

# Proposed solution

- A blockchain-based decentralized energy trading platform is proposed.
- Uses Ethereum blockchain to ensure transparency and security.
- Smart contracts automate energy transactions without intermediaries.
- Each transaction is immutable and recorded on a distributed ledger.
- Users can directly trade surplus energy in real time.



# System architecture

- The system consists of producers, consumers, smart contracts, and blockchain.
- Producers list surplus energy with price and quantity.
- Consumers browse listings and initiate purchase transactions.
- Frontend interacts with blockchain using Web3.js.
- Smart contracts validate and execute transactions automatically.

# Workflow

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- User logs into the system using MetaMask wallet.
- Producer uploads energy listing with required details.
- Consumer selects desired energy listing.
- Smart contract checks conditions like balance and availability.
- Transaction is executed and ownership is transferred.
- Transaction is permanently recorded on blockchain.

# Fututre Scope

- Integration with IoT devices for real-time energy tracking.
- Use of Layer-2 solutions to improve scalability.
- Incorporation of AI for demand prediction and pricing.
- Cross-chain interoperability for broader adoption.
- Alignment with government regulations and policies.

# Conclusion

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- Blockchain enables a secure and decentralized energy trading ecosystem.
- Smart contracts ensure automation and trust.
- System improves efficiency, transparency, and cost-effectiveness.
- Has strong potential for real-world implementation.
- Future improvements can further enhance scalability and adoption.

# Thank You